Lecture 13

k-means → ① centroid-based
② iterative

distance metric

sample $x_n \in \mathbb{R}^d$

$$J(\theta, U) = \sum_{n=1}^{N} \sum_{k=1}^{K} u_{n,k} \, d^2(x_n, \theta_k)$$

for our cluster

$$u_{n,k} = [1, 0, 0, \ldots, 0]$$

cluster membership value for sample n and cluster k

cluster representative or "centroid" $\theta_k \in \mathbb{R}^d$

$$= \sum_{n=1}^{N} \sum_{k=1}^{K} u_{n,k} \|x_n - \theta_k\|_2^2$$

$U \equiv$ membership matrix →
$$\begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ & & \downarrow & & \end{bmatrix}$$

optimization of the objective

$$\arg_{\theta, U} \min J(\theta, U)$$

alternating optimization with EM

E-step: solve for $U \equiv$ memberships

$$J(\theta, U) = \sum_{n=1}^{N} \left(\cancel{\sum_{k=1}^{K}} u_{n,k} \, \| x_n - \theta_k \|_2^2 \right)$$

$$u_{nk} = \begin{cases} 1 & \text{if } k = \arg\min_k \| x_n - \theta_k \|_2^2 \\ 0 & \text{otherwise} \end{cases}$$

M-step — solving for the cluster representatives $\theta_k \in \mathbb{R}^d$

$\mathbb{R}^{1\times d} \times \mathbb{R}^{d\times 1}$

$$J(\theta, U) = \sum_{n=1}^{N}\sum_{k=1}^{K} u_{nk}(x_n - \theta_k)^T(x_n - \theta_k)$$

$$= \sum_{n=1}^{N}\sum_{k=1}^{K} u_{nk}\left(x_n^T x_n - x_n^T\theta_k - \theta_k^T x_n + \theta_k^T\theta_k\right)$$

$$= \sum_{n=1}^{N}\cancel{\sum_{k=1}^{K}} u_{nk}\left(\cancel{x_n^T x_n} - 2\theta_k^T x_n + \theta_k^T\theta_k\right)$$

$$\frac{\partial J}{\partial \theta_k} = \sum_{n=1}^{N} u_{nk}\left(-2x_n + 2\theta_k\right) = 0$$

$$\cancel{2}\sum_{n=1}^{N} u_{nk}(x_n - \theta_k) = 0$$

$$\sum_{n=1}^{N} u_{nk}\theta_k = \sum_{n=1}^{N} u_{nk}x_n$$

$$\theta_k \sum_{n=1}^{N} u_{nk} = \sum_{n=1}^{N} u_{nk}x_n$$

$$\theta_k = \frac{\sum_{n=1}^{N} u_{nk}x_n}{\sum_{n=1}^{N} u_{nk}}$$

indicator

$$\theta_k = \frac{\sum_{n=1}^{N} u_{nk} x_n}{\sum_{n=1}^{N} u_{nk}} \longrightarrow \sum_{n=1}^{N} u_{nk} = N_k$$

$N_k \equiv$ # of samples in cluster C_k

$$= \frac{\sum_{x_n \in C_k} x_n}{N_k}$$

cluster number k columns

$$\underset{N\times K}{U} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ & & & & \end{bmatrix}$$

sample number N rows

$$J(\theta, U) = \sum_{n=1}^{N} \sum_{k=1}^{K} u_{n,k} \|x_n - \theta_k\|_2^2$$

Ideal clusters → spherical
and
vichm to feature
scaling

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softness parameter

$$J(\theta, U) = \sum_{n=1}^{N} \sum_{k=1}^{K} u_{nk}^{m} \, d^2(x_n, \theta_k)$$

$u_{nk} \in \{u_{ij} \in \mathbb{R} \mid 0 \leq u_{ij} \leq 1\}$ → nonnegativity

s.t. $\sum_{k=1}^{K} u_{nk} = 1$ and $0 \leq u_{nk} \leq 1$

sum-to-one constraint

$x_n \in \mathbb{R}^d \rightarrow$ sample

$\theta_k \in \mathbb{R}^d \rightarrow$ cluster representative

Optimization

Cluster Representatives:

$$J = \sum_{n=1}^{N} \sum_{k=1}^{K} u_{nk}^{m} \|x_n - \theta_k\|_2^2$$

$$\frac{\partial J}{\partial \theta_k} = -\cancel{2} \sum_{n=1}^{N} u_{nk}^{m} (x_n - \theta_k) = 0$$

$$\theta_k \sum_{n=1}^{N} u_{nk}^{m} = \sum_{n=1}^{N} u_{n,k}^{m} x_n$$

$$\theta_k = \frac{\sum_{n=1}^{N} u_{nk}^{m} x_n}{\sum_{n=1}^{N} u_{nk}^{m}}$$

$\leftarrow$ this is no longer N_k

membership values :

$U \equiv$ memberships $\quad m > 1$

$$\sum_{k=1}^{K} u_{nk} = 1$$

Lagrangian equivalent to objective :

$$\sum_{n=1}^{N} \sum_{k=1}^{K} u_{nk}^{m} \| x_n - \theta_k \|_2^2 + \sum_{n=1}^{N} \lambda_n \left(1 - \sum_{k=1}^{K} u_{nk} \right)$$

lagrange multipliers

$$\frac{\partial J}{\partial u_{nk}} = m u_{nk}^{m-1} \| x_n - \theta_k \|_2^2 - \lambda_n = 0$$

$$m u_{nk}^{m-1} \| x_n - \theta_k \|_2^2 = \lambda_n$$

$$u_{nk}^{m-1} = \frac{\lambda_n}{m \| x_n - \theta_k \|_2^2}$$

$$u_{nk} = \left(\frac{\lambda_n}{m \| x_n - \theta_k \|_2^2} \right)^{\frac{1}{m-1}}$$

$m \to \infty$

$$\lim_{m \to \infty} u_{nk} = \lim_{m \to \infty} \left(\frac{\lambda_n}{m \| x_n - \theta_k \|_2^2} \right)^{\frac{1}{m-1}}$$

$m \to \infty$

$$\lim_{m\to\infty} u_{nk} = \lim_{m\to\infty} \left(\frac{\lambda_n}{m \| x_n - \theta_k \|_2^2} \right)^{\frac{1}{m-1}}$$

$$\downarrow \left(\frac{\lambda_n}{\infty \| x_n - \theta_k \|_2^2} \right)^{\left(\frac{1}{\infty}\right)} \rightarrow \text{really small}$$

$m \to \infty$

$u_{nk} \Rightarrow$ fuzzier.
and points belong moreso
evenly to every cluster

$m \to 0$

approach the k-means solution

we don't know λ_n ?

$$u_{nk} = \left(\frac{\lambda_n}{m \, \|x_n - \theta_k\|_2^2} \right)^{\frac{1}{m-1}}$$

$\sum_{k=1}^{K} u_{nk} = 1$:

$$\sum_{k=1}^{K} u_{nk} = \sum_{k=1}^{K} \left(\frac{\lambda_n}{m \, \|x_n - \theta_k\|_2^2} \right)^{\frac{1}{m-1}}$$

$$1 = \sum_{k=1}^{K} \left(\frac{\lambda_n^{\frac{1}{m-1}}}{\left(m \, \|x_n - \theta_k\|_2^2\right)^{\frac{1}{m-1}}} \right)$$

$$1 = \lambda_n^{\frac{1}{m-1}} \sum_{k=1}^{K} \frac{1}{\left(m \, \|x_n - \theta_k\|_2^2\right)^{\frac{1}{m-1}}}$$

$$\lambda^{\frac{1}{m-1}} = \left(\sum_{k=1}^{K} \frac{1}{\left(m \, \|x_n - \theta_k\|_2^2\right)^{\frac{1}{m-1}}} \right)^{-1}$$

$$u_{nk} = \left(\frac{\lambda_n}{m \|x_n - \theta_k\|_2^2} \right)^{\frac{1}{m-1}}$$

$$u_{nk} = \left(\sum_{k=1}^{K} \left(\frac{1}{\left(m \|x_n - \theta_k\|_2^2\right)^{\frac{1}{m-1}}} \right) \right)^{-1} * \left(\frac{1}{m \|x_n - \theta_k\|_2^2} \right)^{\frac{1}{m-1}}$$

$$= \frac{1}{\sum_{k=1}^{K} \left(\frac{\|x_n - \theta_k\|_2^2}{\|x_n - \theta_j\|_2^2} \right)^{\frac{1}{m-1}}}$$